Phase 1, Module 4: Operators & Logic Engines

Khyber AI Systems Engineer (KASE)

Welcome to Module 4. An AI is just a massive calculator making logical decisions millions of times per second. Today, we learn the mathematical and logical operators that drive those decisions.

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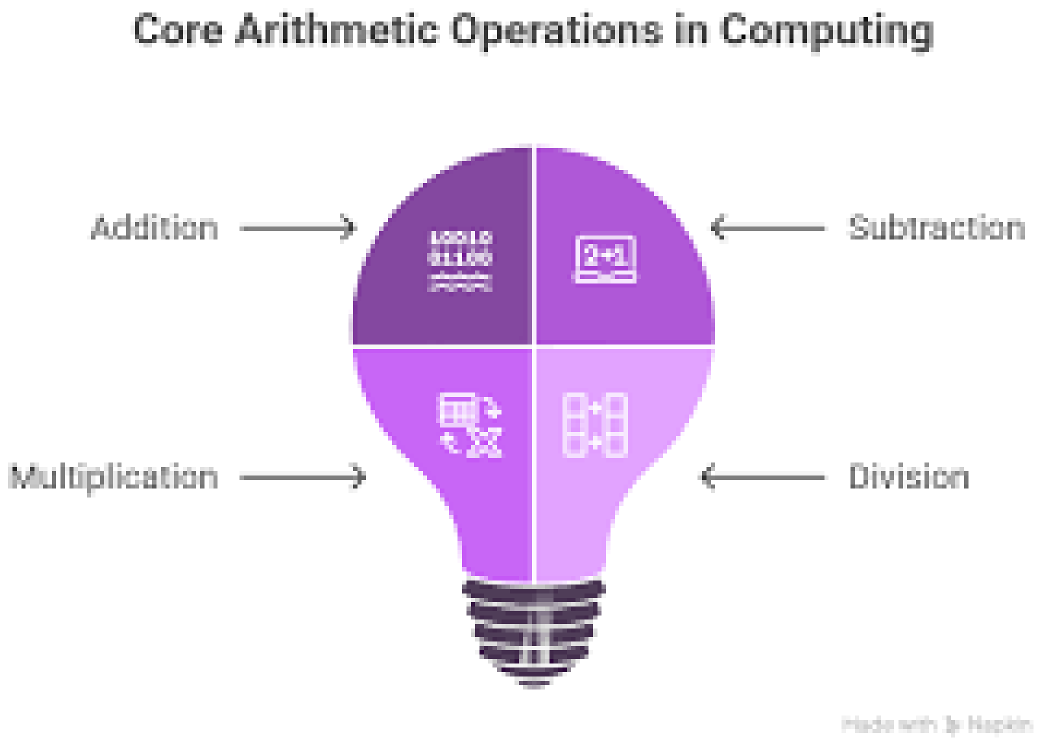
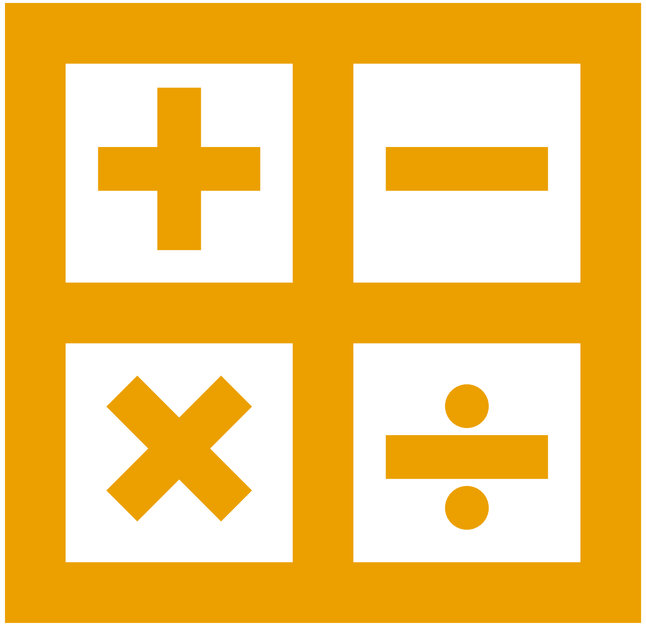
Core Arithmetic (The AI Lens)



Feature Scaling $(+,-,*,/)$

- $+$ / $-$: Used for adjusting AI "Biases" or centering data.
- **:* Used for applying "Weights" to data.
- */:* Crucial for Normalization (e.g., dividing image pixels by 255 so the AI can process them faster).

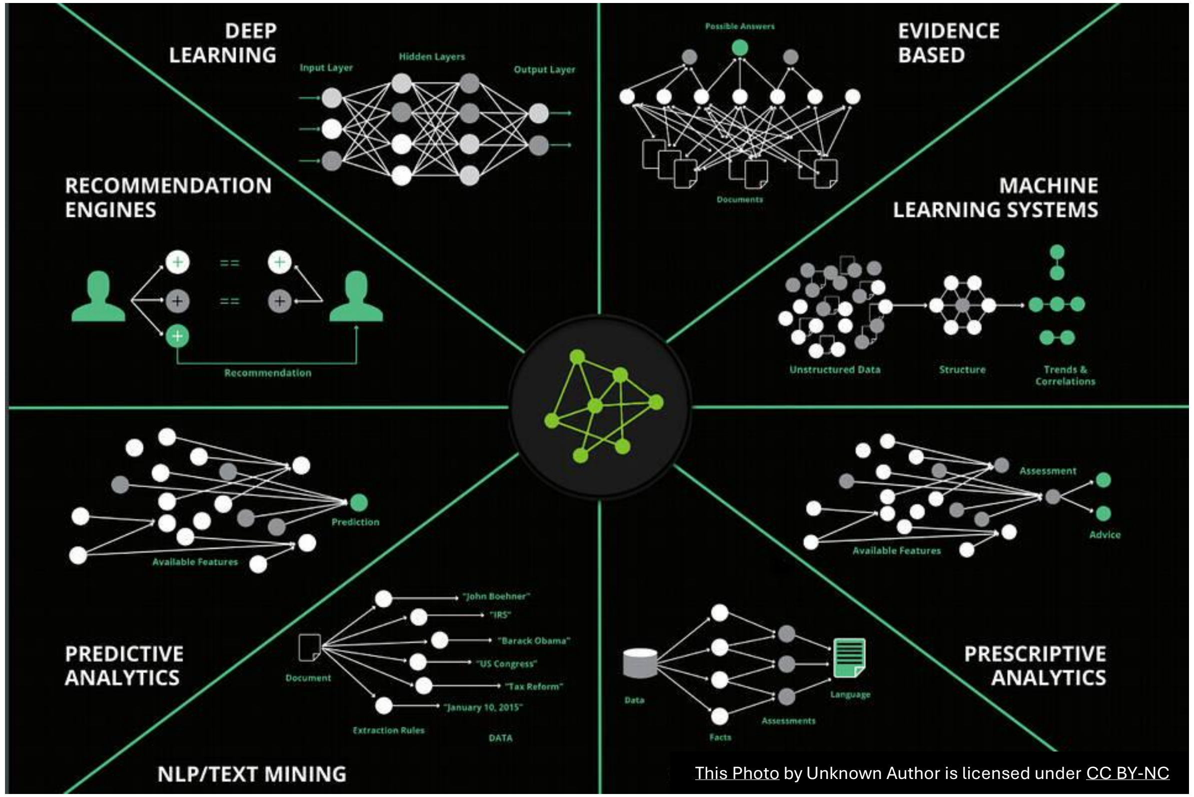
Exponents & The Cost of Being Wrong



Exponentiation $(**)$

- In Python, ^ is NOT an exponent (it's a bitwise operator).
- Use $**$ for powers (e.g., $2 ** 3 = 8$).
- AI Use Case: Used constantly to calculate "Mean Squared Error" — mathematically punishing the AI when it makes a wrong prediction.

The Math of AI (Beyond Addition)



Floor Division & Modulo

- Standard Division (/): Always returns a Float (decimal).
- Floor Division (//): Chops off the decimal. Returns an Integer. (Crucial for splitting data into batches).
- Modulo (%): Returns the remainder of division. (Used to trigger events, like saving an AI model every 10 epochs).

The Interview Trap (== vs *is*)

```
>>> a = [1, 2, 3]
>>> b = a
>>> c = a[:]

>>> a == b
True
>>> a == c
True

>>> a is b
True
>>> a is c
False
```

Value Equality vs. Memory Identity

- ==: Do these two variables hold the exact same value?
- *is*: Are these two variables pointing to the exact same address in RAM?